Fpt University

Web Applications & Web Containers

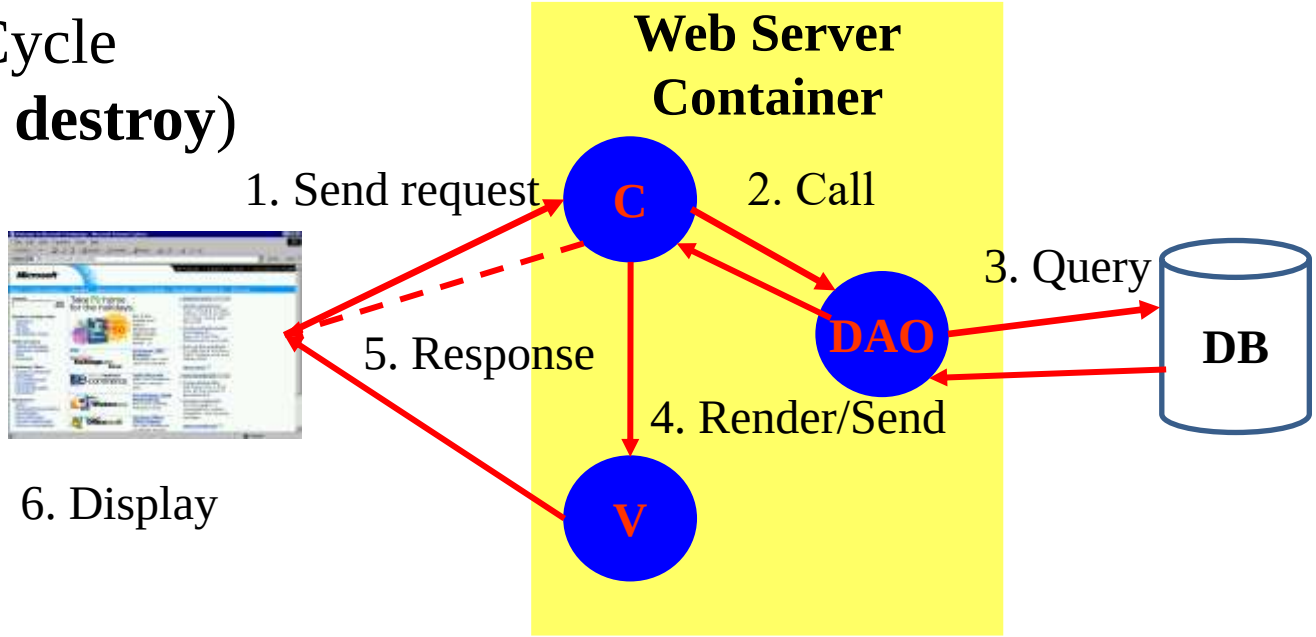
Web Applications
The Web Container Model

#Servlet #Tomcat #Deploy
#Dispatcher #Scope #Video

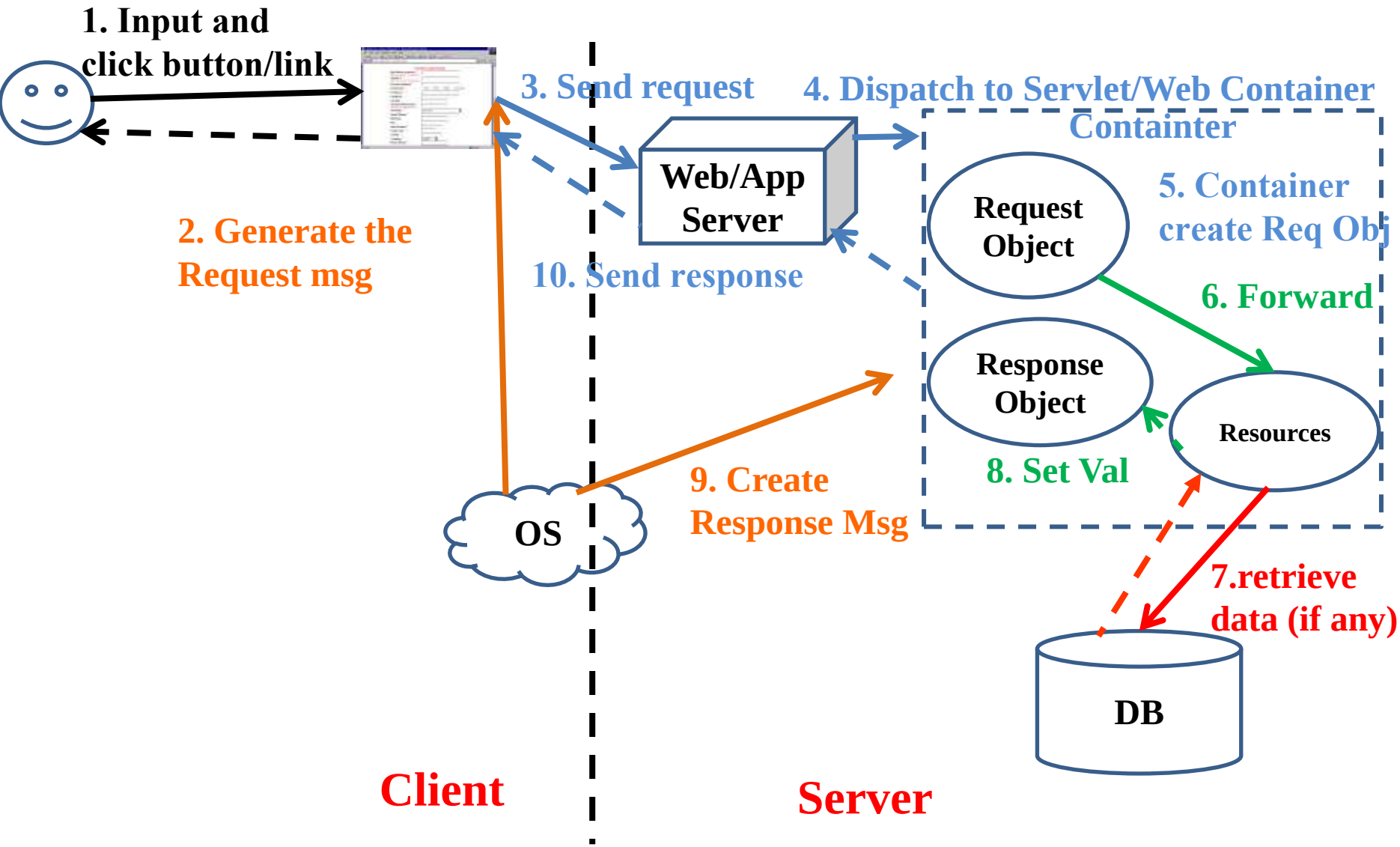
Kiều Trọng Khánh

Review

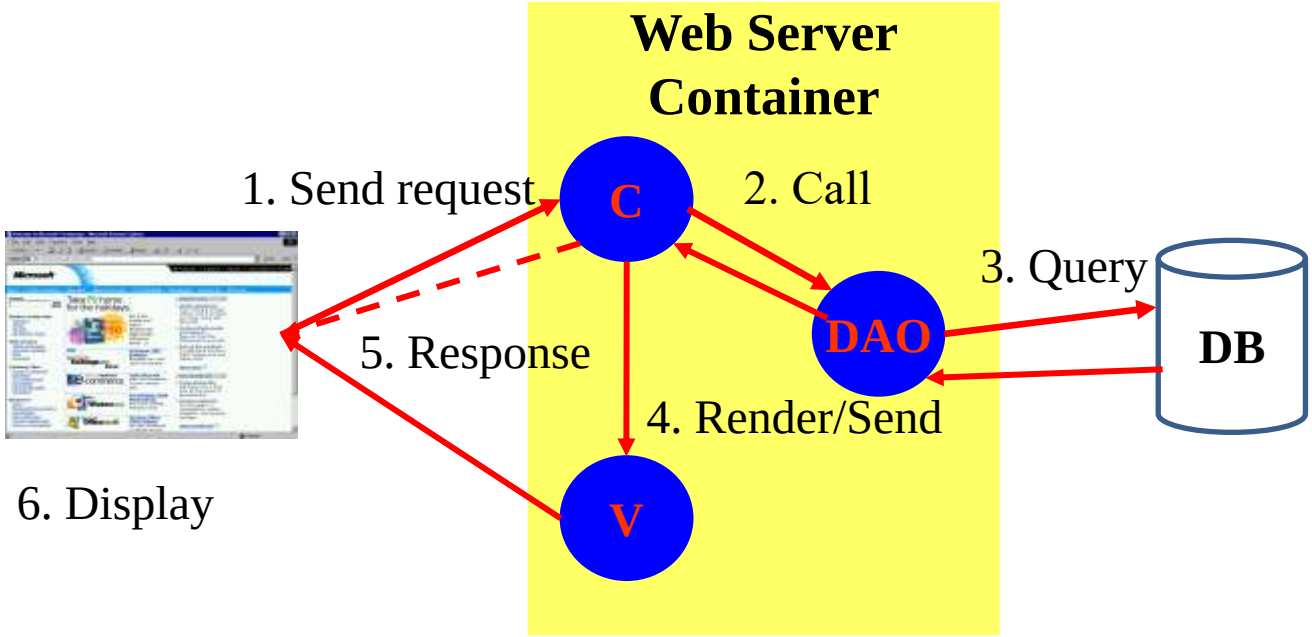
- **How to build the simple web site using html and servlet?**
  - Break down structure component in building web application
- **Some concepts**
  - Servlet vs. Java class, Parameter vs. Variable
  - Form Parameters
  - Http Protocol
  - HTTP Methods: **GET**, **POST**, …
  - Servlet Life Cycle (**init, service, destroy**)
  - JDBC



Review


1. Input and click button/link
2. Generate the Request msg
3. Send request
4. Dispatch to Servlet/Web Container
Containter
Web/App Server
Request Object
5. Container create Req Obj
6. Forward
Response Object
Resources
8. Set Val
9. Create Response Msg
10. Send response
OS
7.retrieve data (if any)
DB
Client
Server

Review


Web Server Container
1. Send request
C
2. Call
3. Query
DAO
DB
5. Response
4. Render/Send
V
6. Display

Objectives

- **How to deploy the Web Application to Web Server without using Netbeans/ Eclipse tools?**
  - Web applications Structure
  - Request Parameters vs. Context Parameters vs. Config/Servlet Parameters
  - Application Segments vs. Scope
- **How to transfer from resources to others with/without data/objects?**
  - Attributes vs. Parameters vs. Variables
  - Redirect vs. RequestDispatcher

Objectives

Web based App - PBL

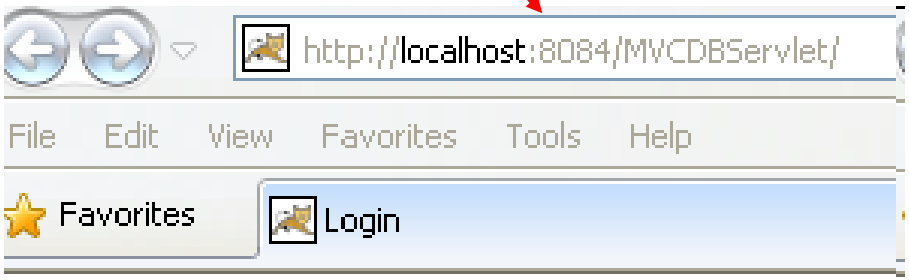
Web

- Day 1, 2, 3, 4, 5, 6 – Login Servlet, JDBC ✓
- Day 7, 8, 9 – Search Break Down
- Day 10, 11 – MVC2 JSP
- Day 12, 13, 14, 15 – CUD, Shopping Carts Sessions
- Day 16, 17 – Login JavaBeans
- Day 18, 19, 20 – CRUD MVC 2 Complete - JSTL - Taglib
- Day 21, 22, 23 – Filter MVC2 Using Filter as Controller

Deploy Application

Expectation

Typed and Press Enter



Login Page

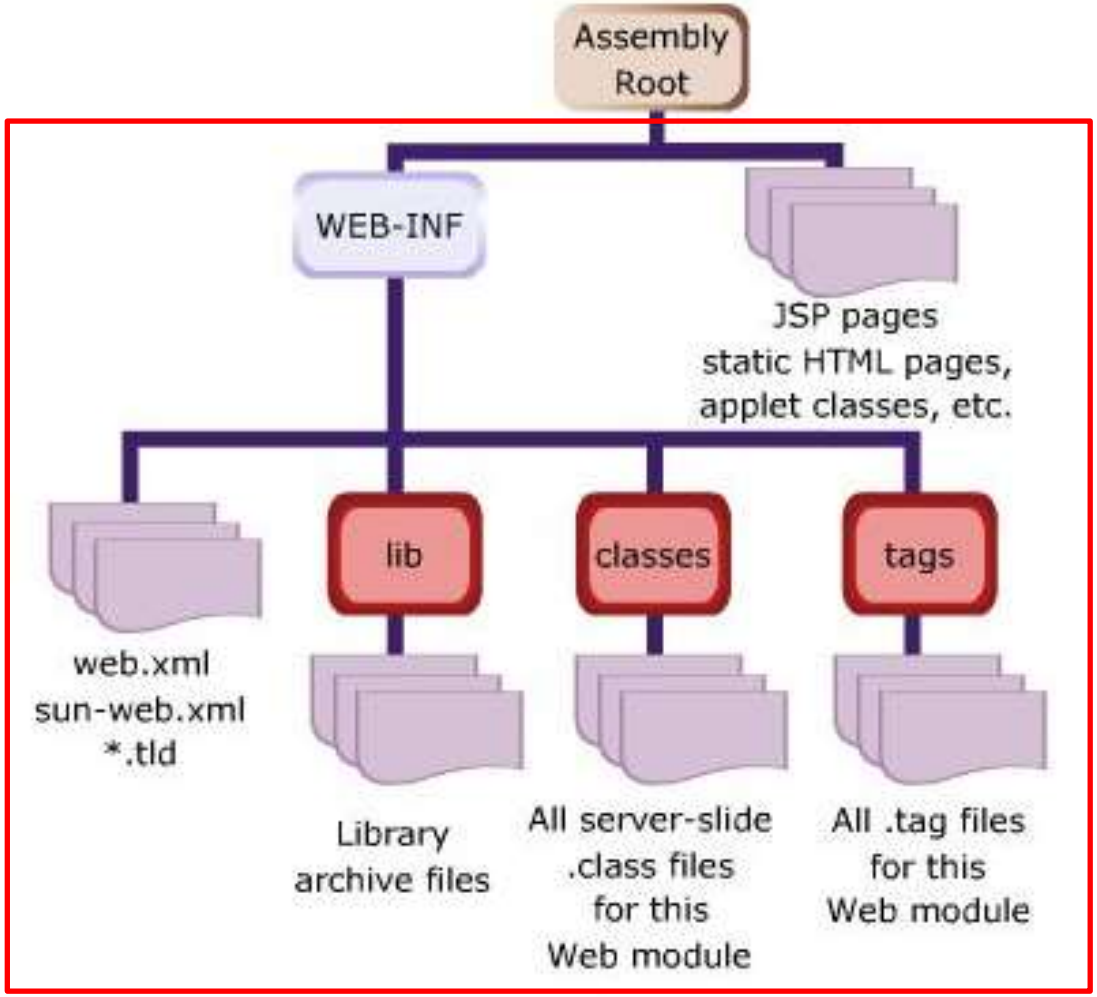
Username

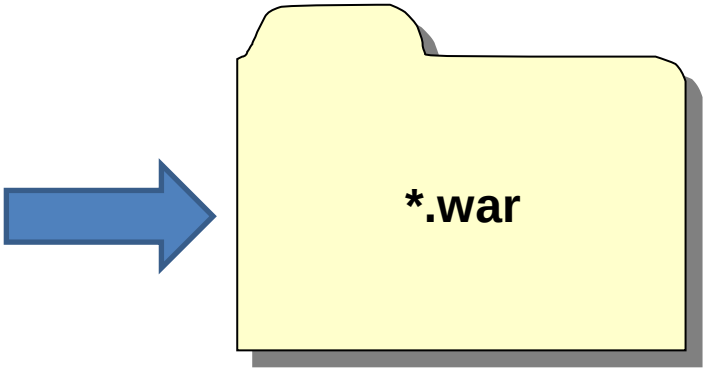
Password

Login Reset

Web Applications

File and Directory Structure





Above structure is packaged into *.war (Web (Application) ARchive) file to deploy on Web Server

Web Applications

File and Directory Structure

- **/WEB-INF/classes** – for **classes that exist** as separate Java classes (*not* packaged within JAR files). These might be servlets or other support classes.
- **/WEB-INF/lib** – for JAR file. These can contain anything at all – the main servlets for your application, supporting classes that connect to databases – whatever.
- **/WEB-INF** itself is the home for an absolutely crucial file called **web.xml**, **the web deployment descriptor** file.
- **2 special rules** apply to files within the /WEB-INF directory
  - Direct client access should be disallowed with an HTTP 404 code
  - The **order** of class **loading** the java classes in the **/WEB-INF/classes** directory should be **loaded before** classes resident in **jar files** in the /WEB-INF/lib directory

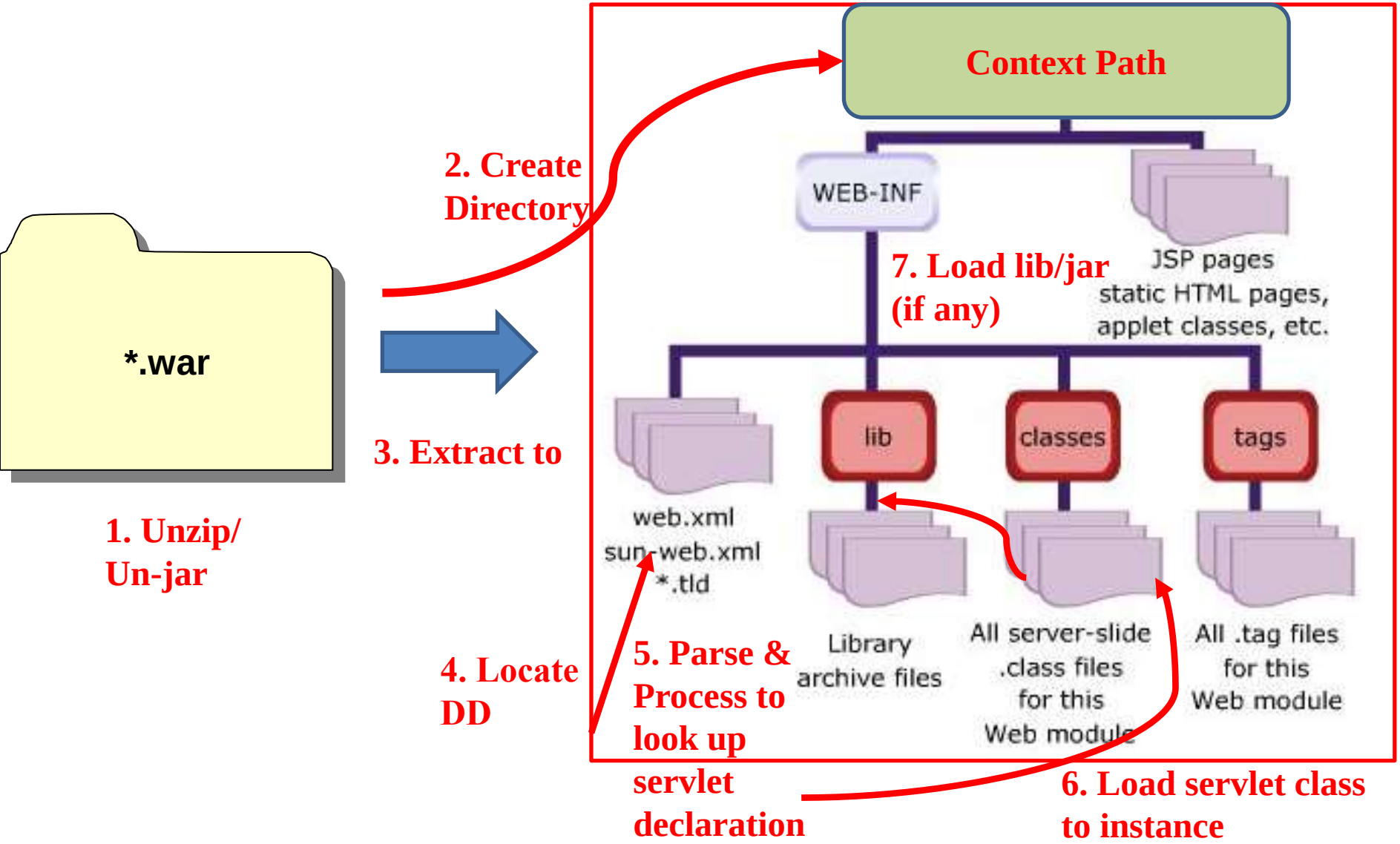
Web Applications

File and Directory Structure

- A Place for Everything and Everything in Its Place.
  - On Tomcat Server, it locates at **CATALINA_HOME/webapps**
  - **Execute**: http://host:port/webappcontext/resourceIneed
- Construct the file and directory structure of a Web **Application** that may **contain**:
  - Static content,
  - JSP pages,
  - Servlet classes,
  - The deployment descriptor,
  - Tag libraries,
  - JAR files and Java class files;
  - and describe how to protect resource file from HTTP access.

Web Applications

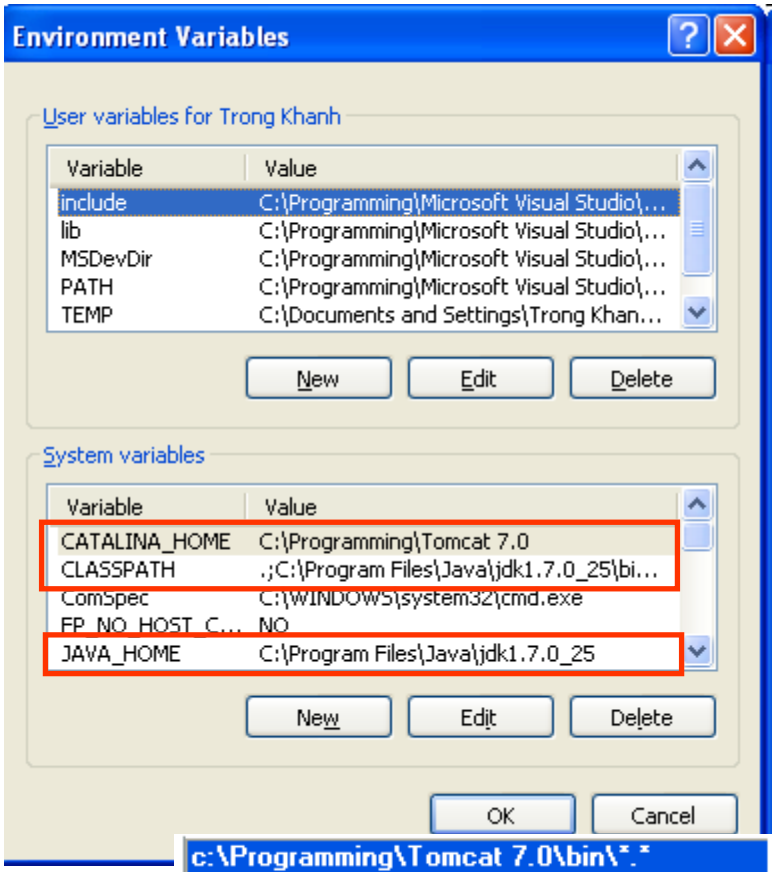
Deploy Mechanism

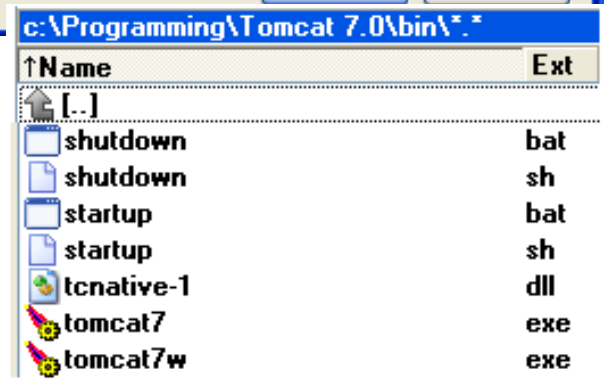


Web Applications

Manual Deploying

- Setup the environment for JAVA and TOMCAT
  - **In Windows OS**: click Properties of Computer, choose "Advanced System Setting", choose Advanced, Click "Environment Variables", to set following environment variables
- Go to the **Installed_Tomcat\bin** directory, click **startup.bat** or **tomcat7w.exe**





Web Applications

Manual Deploying

Tomcat

```
INFO: Deploying web application directory C:\Programming\Tomcat 7.0\webapps\docs

thg 9 21, 2013 8:27:33 CH org.apache.catalina.startup.HostConfig deployDirectory

INFO: Deploying web application directory C:\Programming\Tomcat 7.0\webapps\exam
ples
thg 9 21, 2013 8:27:33 CH org.apache.catalina.startup.HostConfig deployDirectory

INFO: Deploying web application directory C:\Programming\Tomcat 7.0\webapps\host
-manager
thg 9 21, 2013 8:27:33 CH org.apache.catalina.startup.HostConfig deployDirectory

INFO: Deploying web application directory C:\Programming\Tomcat 7.0\webapps\mana
ger
thg 9 21, 2013 8:27:33 CH org.apache.catalina.startup.HostConfig deployDirectory

INFO: Deploying web application directory C:\Programming\Tomcat 7.0\webapps\ROOT

thg 9 21, 2013 8:27:33 CH org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["http-apr-8080"]
thg 9 21, 2013 8:27:33 CH org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["ajp-apr-8009"]
thg 9 21, 2013 8:27:33 CH org.apache.catalina.startup.Catalina start
INFO: Server startup in 619 ms
```

c:\Programming\Tomcat 7.0\webapps*.*

Name	Ext	Size
[..]		<DIR>
[AJDay1_7]		<DIR>
[docs]		<DIR>
[examples]		<DIR>
[host-manager]		<DIR>
[manager]		<DIR>
[ROOT]		<DIR>
AJDay1_7	war	25.9

c:\Programming\Tomcat 7.0*.*

Name	Ext	Size
[..]		<DIR
[bin]		<DIR
[conf]		<DIR
[lib]		<DIR>
[logs]		<DIR>
[temp]		<DIR>
[webapps]		<DIR>
[work]		<DIR>
LICENSE		57.862
NOTICE		1.228
RELEASE-NOTES		9.054
RUNNING	txt	16.742

Tomcat

```
INFO: Server startup in 619 ms
thg 9 21, 2013 8:29:03 CH org.apache.catalina.startup.HostConfig deployWAR
INFO: Deploying web application archive C:\Programming\Tomcat 7.0\webapps\AJDay1
_7.war
```

- Testing on web browser
- Delete the war file and the directory to undeploy application
- Press Ctrl + C to stop server

The Web Container Model

The Servlet Container

- Is a **compiler,** executable program.
- Is the **intermediary** between the Web server and the servlets in the container.
- **Loads, initializes, and executes** the servlets.
  - When a **request arrives**, the container **maps** the **request to** a **servlet**, **translates** the **request**, and then **passes** the **request** to the servlet.
  - The servlet **processes** the **request** and **produces** a **response**.
  - The container **translates** the **response** into the **network format**, then **sends** the response **back** to the Web server.
- Is designed to perform well while **serving large** numbers of **requests**.
- Can hold any number of active servlets, filters, and listeners.
- Both the container and the objects in the container are **multithreaded**.
  - The container creates and manages threads as necessary to handle incoming requests.
  - The container handles multiple requests concurrently, and more than one thread may enter an object at a time.
  - Therefore, each object within a container must be threadsafe.

The Web Container Model

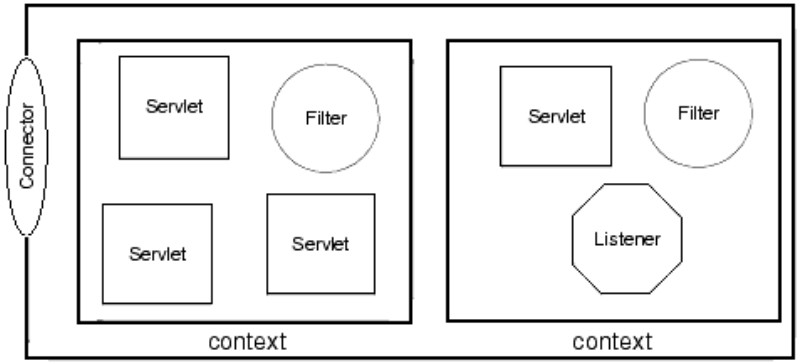
The Servlet Container

- Fortunately,
  - We are a web *component* developer, not a *web container* developer.
  - So we can take for granted much of what is built into the web container.
- We are a **consumer** of what the web container provides, and
- We have to understand the infrastructure only insofar as it affects our own business applications

The Web Container Model

The ServletContext

- Is considered as a **memory segment** that
  - **Collects all methods that are used for particular Web application in server side**
  - **Support to interact with Servlet container**
  - **Stores some object in server side that all web's component can access**
  - **Exists from the application has been deployed to undeployed (or server is crashed)**
- The container uses a *context* to
  - **Group related** components.
  - **Share** data in easily.
  - **Provide** a set of **services** for the web application to work with the container
- **Each context** usually corresponds to a **distinct** Web application.



The Web Container Model

The ServletContext – Example

- The directory structure below describes **two contexts**, one named **day1** and one named **day2**. The day2 context contains a static HTML page, intro.html.

webapps

\day1

\WEB-INF

web.xml

\day2

intro.html

\WEB-INF

web.xml

The Web Container Model

The ServletContext – Initialization Parameters

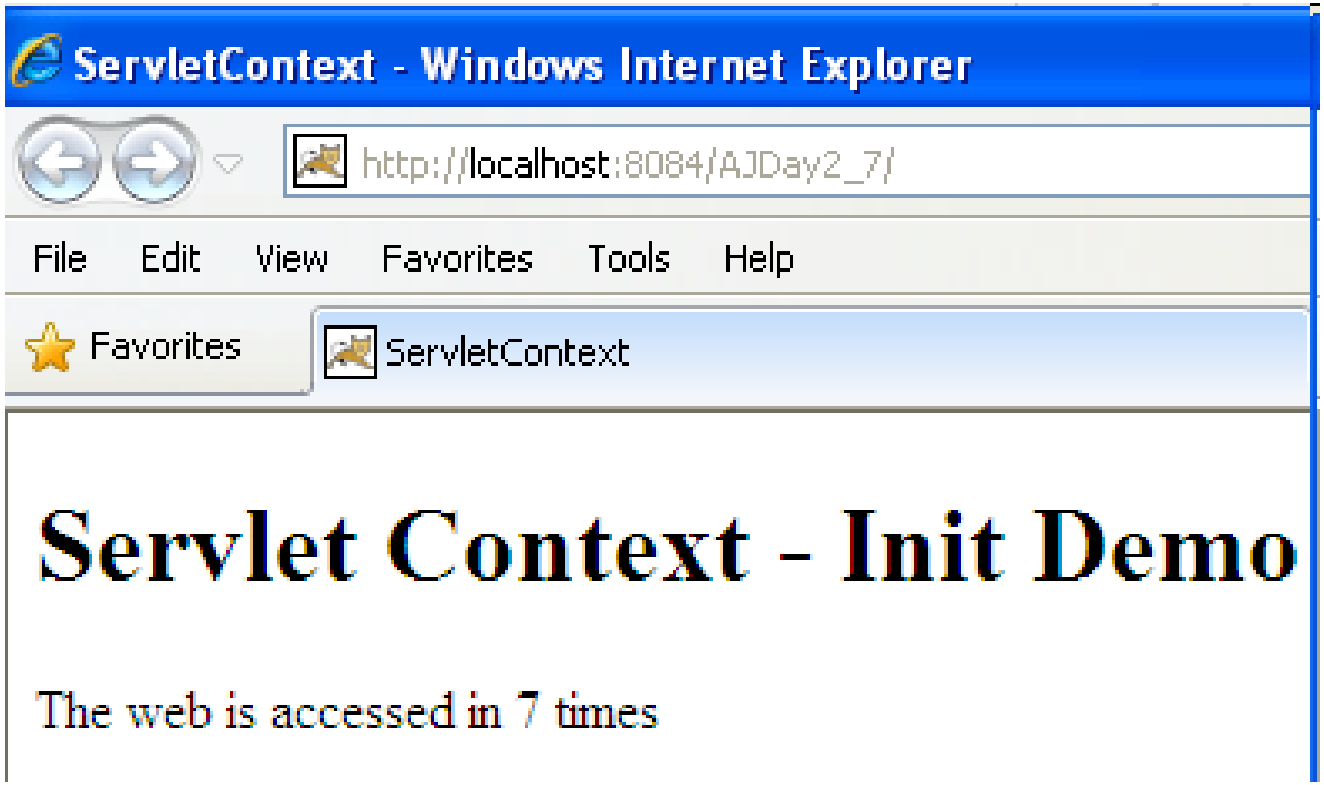
- Providing some fundamental information available to all the dynamic resources (servlets, JSP) within the web application is allowed by
  - **Using** servlet **initialization parameters in** the **deployment descriptor** with the **getInitParameter(String parName)** method to provide **initialization information for servlets**
  - The servlet initialization parameters is accessible only from its containing servlet
- Setting up the Deployment Descriptor

```
<web-app>
    <context-param>
            <param-name>parName</param-name>
            <param-value>parValue</param-value
    </context-param>
   …
</web-app>
```

The Web Container Model

The ServletContext – Initialization Parameters

- Example
  - Building the web application have the counter function that allows the web site can account the number of accessed users
  - The application's GUI should be same as

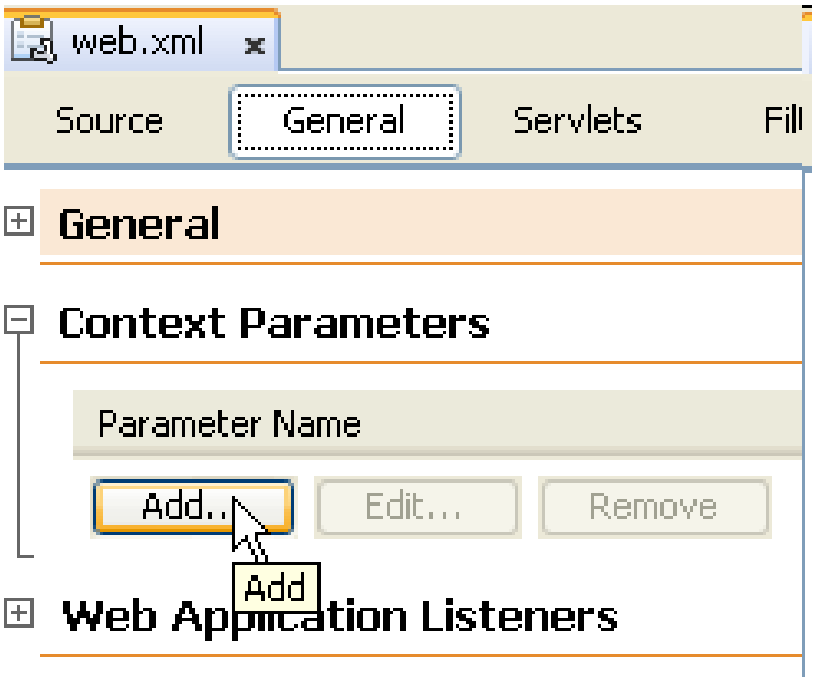


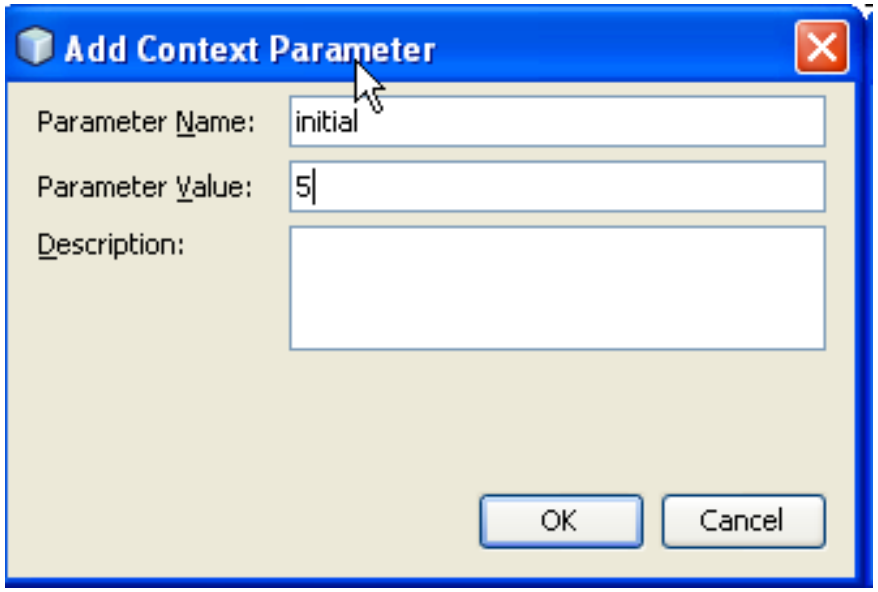
The Web Container Model

The ServletContext – Initialization Parameters

- Writing Code to Retrieve ServletContext Initialization Parameters

```
ServletContext sc = getServletContext();
String var = sc.getInitParameter("parName");
```





The Web Container Model

The ServletContext – Initialization Parameters

web.xml

Source | General | Servlets | Filters | Pages | Re

General

Context Parameters

Parameter Name	Parameter Value
initial	5

Add... | Edit... | Remove

web.xml

Source | General | Servlets | Filters | Pages | Refe

```
1  <?xml version="1.0" encoding="UTF-8"?>
2  <web-app version="2.5" xmlns="http://java
3      <context-param>
4          <param-name>initial</param-name>
5          <param-value>5</param-value>
6      </context-param>
```

web.xml | InitCounter.java

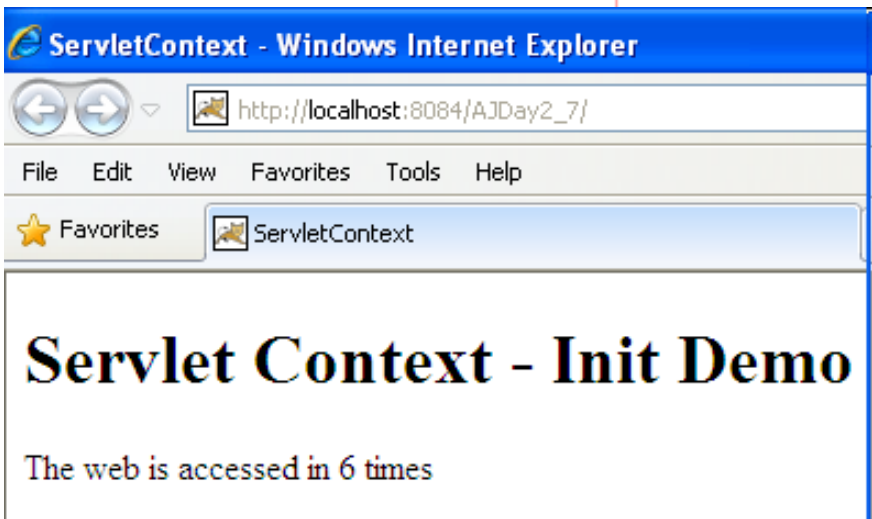
Source | History

```
16      *
17      * @author Trong Khanh
18      */
19     public class InitCounter extends HttpServlet {
20         private int count;
21         public void init() throws ServletException {
22             super.init();
23             ServletContext sc = getServletContext();
24             String tmp = sc.getInitParameter("initial");
25             try {
26                 count = Integer.parseInt(tmp);
27             } catch (NumberFormatException e) {
                   e.printStackTrace();
29             }
30         }
```

The Web Container Model

The ServletContext – Initialization Parameters

```
protected void processRequest(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    PrintWriter out = response.getWriter();
    try {
        ...
        out.println("<body>");
        out.println("<h1>The ServletContext-Init Demo</h1>");
        count++;
        out.println("The web is accessed in " + count + "times");
        ...
        out.println("</body>");
        out.println("</html>");
    } finally {
        out.close();
    }
}
```



The Web Container Model

The ServletContext – **Initialization Parameters**

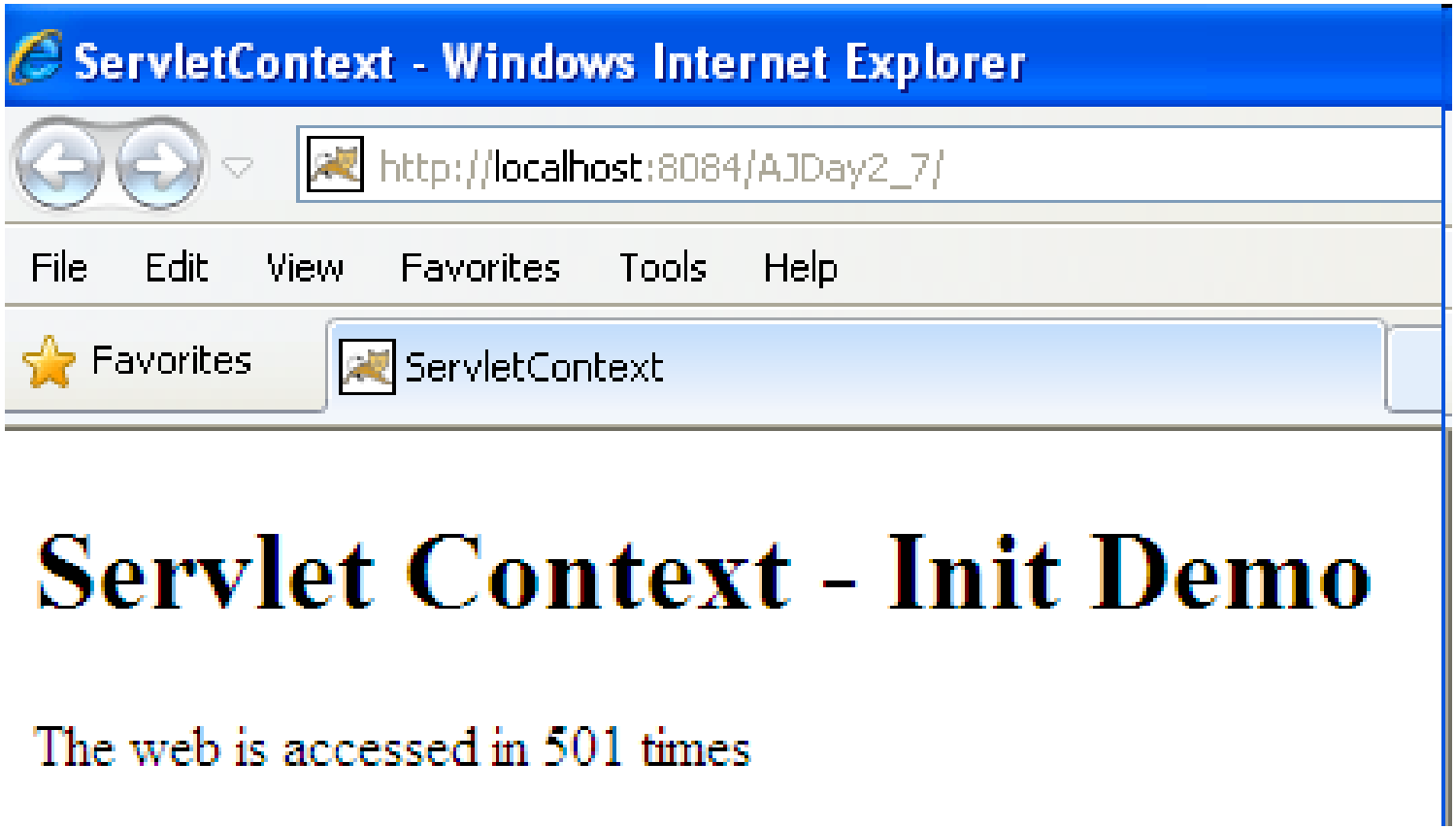
web.xml

Source | General | Servlets | Filters | Pages | Refer

```
1   <?xml version="1.0" encoding="UTF-8"?>
2   <web-app version="2.5" xmlns="http://java.
3       <context-param>
4           <param-name>initial</param-name>
5           <param-value>5</param-value>
6       </context-param>
7       <context-param>
8           <param-name>initial</param-name>
9           <param-value>500</param-value>
10      </context-param>
11      <servlet>
```

The Web Container Model

The ServletContext – **Initialization Parameters**



The Web Container Model

The ServletConfig interface

- To **pass as an argument** during initialization, the servlet container uses an object of ServletConfig interface
- **Configuring a servlet before processing** requested data
- Retrieve servlet initialization parameters

Methods	Descriptions
getServletName	- **public String getServletName()** - Searches the configuration information and retrieves name of the servlet instance - String servletName = getServletName();
getInitParameter	- **public String getInitParameter (String name)** - Retrieves the value of the initialisation parameter - Returns null if the specified parameter does not exist - String password = getInitParameter(”password”);
getServletContext	- **public ServletContext getServletContext()** - returns a ServletContext object used by the servlet to interact with its container. - ServletContext ctx = getServletContext();

The Web Container Model

The ServletConfig – Initialization Parameters

- Setting up the Deployment Descriptor

```
<servlet>
    <servlet-name>servletName</servlet-name>
    <servlet-class>servletClass</servlet-class>
      <init-param>
        <param-name>parName</param-name>
        <param-value>parValue</param-value>
      </init-param>
</servlet>
```

- Writing Code to Retrieve ServletConfig Initialization Parameters

```
ServletConfig sc = getServletConfig();
String name = sc.getInitParameter("parName");
```

The Web Container Model

The ServletConfig interface – Example

The Web Container Model

The ServletConfig interface – Example

web.xml | InitConfigCounter.java

Source | General | Servlets | Filters | Pages | References | Security

Servlets

InitCounter -> /InitCounter

InitConfigCounter -> /InitConfigCounter

Servlet Name: InitConfigCounter Startup

Description:

Servlet Class: sample.servlet.InitConfigCounter Browse... Go to

JSP File: Browse... Go to

URL Pattern(s): /InitConfigCounter

Use comma (,) to separate multiple patterns.

Initialization Parameters:

Parameter Name	Parameter Value	Description

Add... | Edit... | Remove

Add

Security Role References:

Add Initialization Parameter

Parameter Name: initial

Parameter Value: 5

Description: init parameter

OK | Cancel

The Web Container Model

The ServletConfig interface – Example

web.xml | InitConfigCounter.java

Source | General | Servlets | Filters | Pages | References | Security | History

```
2  <web-app version="2.5" xmlns="http://java.sun.com/xml/ns/javaee" xmlns::
3      <context-param>
7      <context-param>
11     <servlet>
15     <servlet>
16         <servlet-name>InitConfigCounter</servlet-name>
17         <servlet-class>sample.servlet.InitConfigCounter</servlet-class>
18         <init-param>
19             <description>init parameter</description>
20             <param-name>initial</param-name>
21             <param-value>5</param-value>
22         </init-param>
23     </servlet>
24     <servlet-mapping>
```

The Web Container Model

The ServletConfig interface – Example

web.xml | InitConfigCounter.java

Source | History

```
17      * @author Trong Khanh
18      */
19     public class InitConfigCounter extends HttpServlet {
20         private int count;
21         public void init(ServletConfig config) throws ServletException {
22             super.init(config);
23             ServletConfig scc = getServletConfig();
24             String tmp = scc.getInitParameter("initial");
25             try {
26                 count = Integer.parseInt(tmp);
27             } catch (NumberFormatException e) {
28                 e.printStackTrace();
29             }
30         }
```

The Web Container Model

The ServletConfig interface – Example

```
42      protected void processRequest(HttpServletRequest request, HttpServle
43              throws ServletException, IOException {
44          response.setContentType("text/html;charset=UTF-8");
45          PrintWriter out = response.getWriter();
46          try {
47              /* TODO output your page here. You may use following sample
48              out.println("<!DOCTYPE html>");
49              out.println("<html>");
50              out.println("<head>");
51              out.println("<title>ServletConfig</title>");
52              out.println("</head>");
53              out.println("<body>");
54              out.println("<h1>Servlet Config - Init Counter Demo</h1>");
55
56              count++;
57              out.println("The web is accessed in " + count + "times");
58              out.println("</body>");
59              out.println("</html>");
60          } finally {
61              out.close();
62          }
63      }
```

The Web Container Model

The ServletConfig interface – Example

```
15  <servlet>
16      <servlet-name>InitConfigCounter</servlet-name>
17      <servlet-class>sample.servlet.InitConfigCounter</servlet-class>
18      <init-param>
19          <description>init parameter</description>
20          <param-name>initial</param-name>
21          <param-value>5</param-value>
22      </init-param>
23      <init-param>
24          <description>init parameter</description>
25          <param-name>initial</param-name>
26          <param-value>5000</param-value>
27      </init-param>
28  </servlet>
29  <servlet-mapping>
```

How To Transfer

Requirements

- After built the web application in the first topic
  - **The search page** allows user **search appropriate the last name of users**
  - **The result** of searching is **shown in the data grid**. In each row, the **information about ordinary number, username, password, last name and roles is shown**
- The GUI of web application is present as following

How To Transfer

Expectation

Home - Windows Internet Explorer

http://localhost:8084/MVCDBServlet/ProcessServlet

File Edit View Favorites Tools Help

Favorites Home

Welcome to DB Servlet

Name a

Search

http://localhost:8084/MVCDBServlet/ProcessServlet?txtName=a&btAction=Search

Edit View Favorites Tools Help

Favorites Search Result

Welcome to DB Servlet

Name a

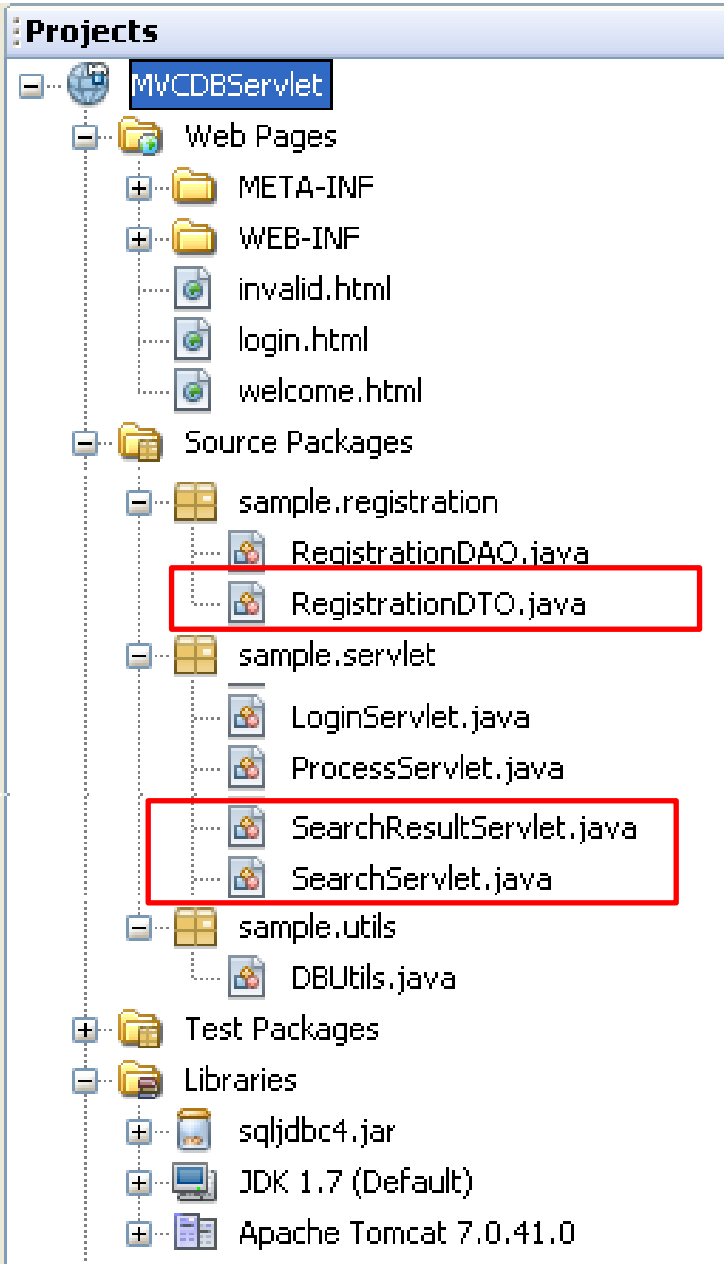
Search

No.	Username	Password	Lastname	Roles	Delete
1	class861	3455454	Khoa	false	Delete
2	EntityClass	EJB32	Day11	true	Delete
3	khanh	kieu123	Kieu Khanh	true	Delete
4	quangabc	123456	Quang di tre	false	Delete
5	Taidaica	123456	Tai Dai Dai Ca	false	Delete
6	TienDan	12345634	Dan Quan Dai	true	Delete

http://localhost:8084/MVCDBServlet/ProcessServlet?btAction=del&txtUsername=khanh&searchValue=a

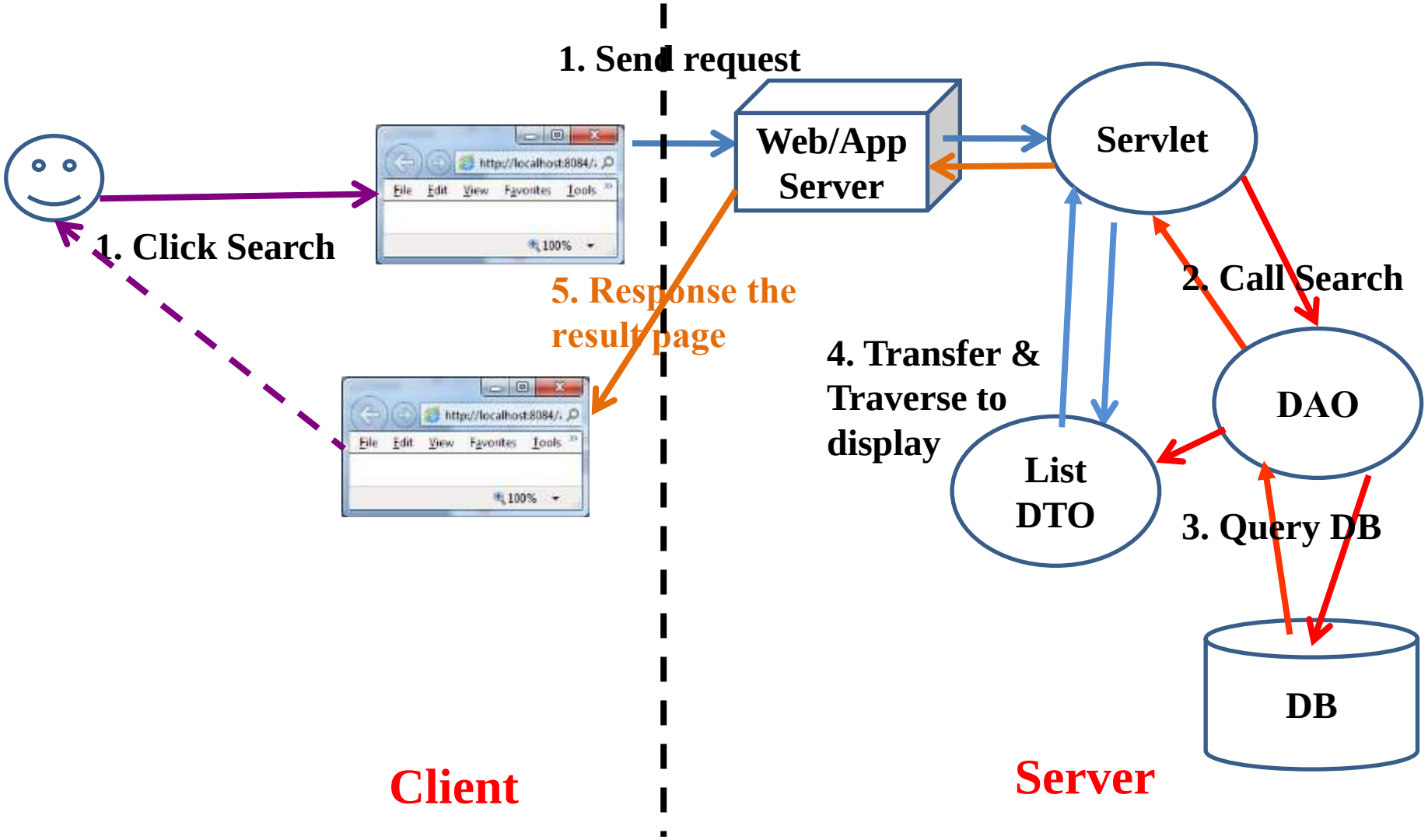
How To Transfer

Expectation



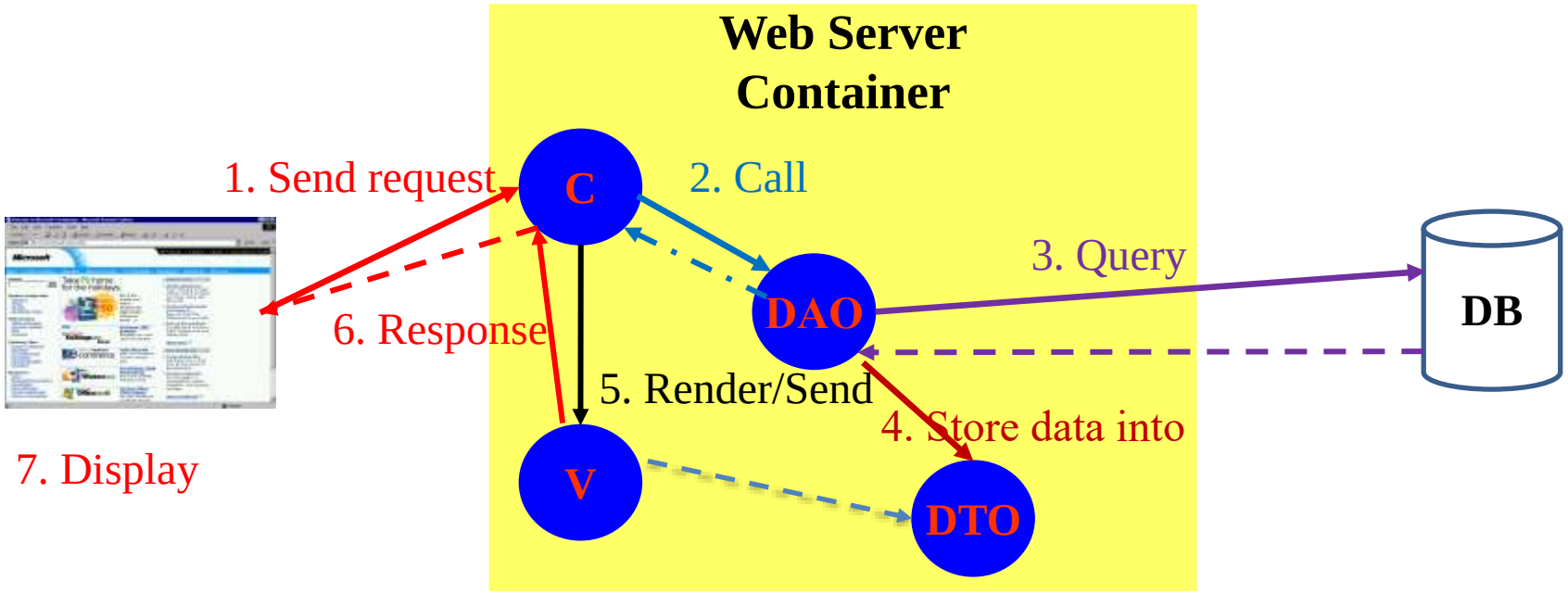
How To Transfer

Interactive Server Model



How To Transfer

Abstraction



The Web Container Model

Need for using attributes

- **Problems:**
  - **How** to **remember** an **user** that has **already logged** into the **particular website**?
  - How to **store a collection** of **selected products online** when the user has **already chosen** while the HTTP is a stateless protocol? Besides, they can search and choose other products
- **Solutions:**
  - **Store** data or object **as long as** user **still browses** the web site
  - Attributes is a qualified candidate: **Attributes** are a **collection of <attribute-name, value> pairs** that is **stored** in a **scope (segment) in server**
  - **Life cycle of them** is **long as** its **defined scope**.

The Web Container Model

Attributes, **Scope**, and Multithreading

- Defines **how long** a reserved **memory segment is available in the context on the server**.
- There are **3 scopes**
  - **Request Scope**
    - **Lasts** from HTTP **request** hits a **web container to** the servlet **deliveres** the HTTP **response**.
    - **javax.servlet.ServletRequest**
  - **Session Scope**
    - **A browser window establishes** up to the point where that **browser window is closed**
    - **Open session** up to the point where that **session is closed, session is time out, server is crashed**.
    - **javax.servlet.http.HttpSession**
  - **Context (Application) Scope**
    - Is the **longest-lived** of the three scopes available to you.
    - Exists **until** the **web container is stopped**.
    - **javax.servlet.ServletContext**

The Web Container Model

Attributes, **Scope**, and Multithreading

- **Choosing Scopes**
  - **Request Scope:** attributes are required for a one-off web page and aren't part of a longer transaction
  - **Session Scope**: attributes are part of a longer transaction, or are spanned **several request** but they are information **unique to particular client**
    - **Ex**: username or account
  - **Context Scope**: attributes can allow **any web resource to access** (e.g. public variables in application)

The Web Container Model

Attributes, Scope, and Multithreading

- **Parameters vs. Attributes**
  - **Parameters** allow information to flow into a web application (**passed** to web application **via form or query string**). They **exist** in **request scope**
  - **Attributes** are more of a means of handling information *within* the web application. They can be **shared or accessed within** their **defined scope**
  - **Data types of Parameter is String but the Attribute is Object**
- The web container uses attributes as a place to
  - **Provide information to interested code**: the way supplement the standard APIs that yield information about the web container
  - **Hang on to information that your application, session, or even request requires later**.
- The developer **can access** the **attribute value with attribute's name**

The Web Container Model

Attributes, Scope, and Multithreading

Methods	Descriptions
getAttribute	- **public Object getAttribute(String name)** - returns the value of the name attribute as Object - Ex: String user = (String)servletContext.getAttribute("USER");
setAttribute	- **public void setAttribute(String name, Object obj)** - Binds an object to a given attribute name in the scope - Replace the attribute with new attribute, if the name specified is already used - servletContext.setAttribute("USER", "Aptech");
removeAttribute	- **public void removeAttribute(String name)** - Removes the name attributes - Ex: servletContext.removeAttribute("USER");
getAttributeNames	- **public Enumeration getAttributeNames()** - Returns an Enumeration containing the name of available attributes. Returns an empty if no attributes exist.

The Web Container Model

Attributes, Scope, and **Multithreading**

- **Multithreading and Request Attributes**
  - request attributes are thread safe (*because everything will only ever be accessed by one thread and one thread alone*)
- **Multithreading and Session Attributes**
  - session attributes are *officially* not thread safe.
- **Multithreading and Context Attributes**
  - context attributes are not thread safe
  - You have **two approaches** to **solve** the multithreading dilemma:
    - **Set up servlet context attributes** in the **init() method** of a servlet that loads on the startup of the server, and at no other time. Thereafter, treat these **attributes as "read only".**
    - If there are **context attributes** where you have no option but to update them later, surround the updates with synchronization blocks.

The Web Container Model

Need for using RequestDispatcher – Redirect

web.xml | requestDispatcher.html

Source | History

```html
<!DOCTYPE html>
<html>
    <head>
        <title>Demo</title>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    </head>
    <body>
        <h1>Demo Request Dispatcher</h1>
        <form action="MiddleServlet">
            Name <input type="text" name="txtName" value="" /><br/>
            <input type="submit" value="Transfer" />
        </form>
    </body>
</html>
```

MiddleServlet

```java
out.println("<h1>Middle Servlet</h1>");

request.setAttribute("Middle", "Middle Information");
response.sendRedirect("EndServlet");

out.println("</body>");
out.println("</html>");
```

The Web Container Model

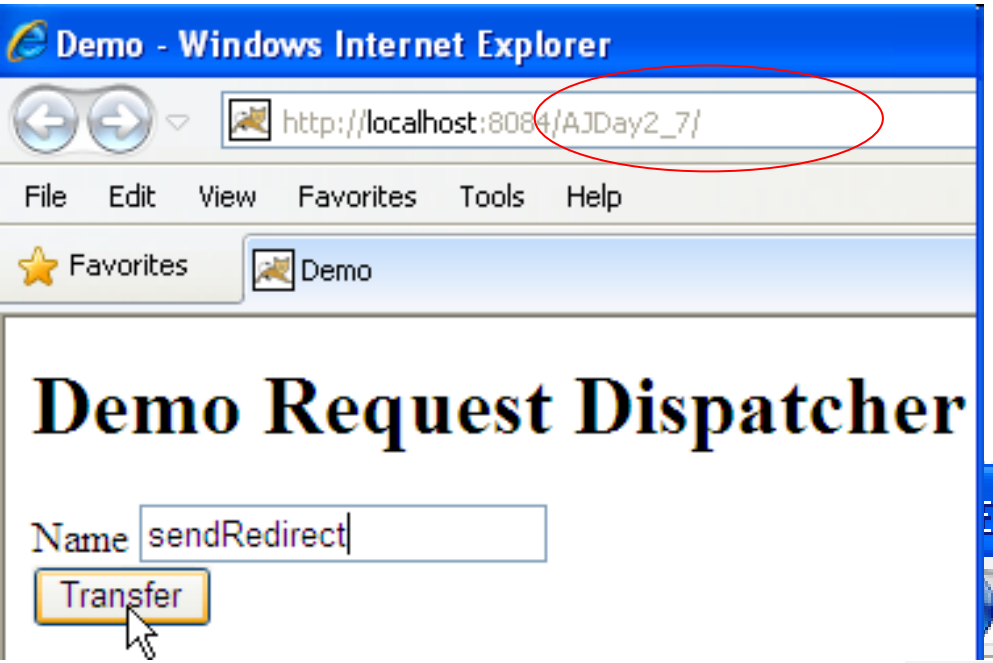
Need for using RequestDispatcher – Redirect

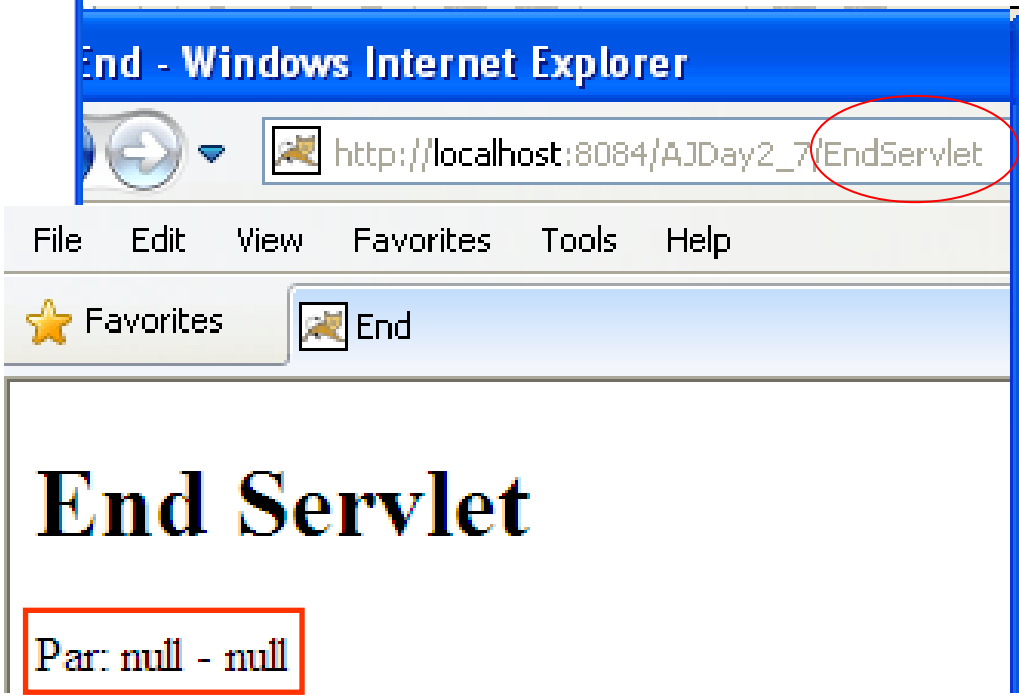
MiddleServlet.java | EndServlet.java

```
28      protected void processRequest(HttpServletRequest request, HttpSe
29      throws ServletException, IOException {
30          response.setContentType("text/html;charset=UTF-8");
31          PrintWriter out = response.getWriter();
32          try {
33
34              out.println("<html>");
35              out.println("<head>");
36              out.println("<title>End</title>");
37              out.println("</head>");
38              out.println("<body>");
39              out.println("<h1>End Servlet</h1>");
40
41              String name = request.getParameter("txtName");
42              String middle = (String)request.getAttribute("Middle");
43
44              out.println("Par: " + name + " - " + middle);
45
46              out.println("</body>");
47              out.println("</html>");
48
49          } finally {
50              out.close();
51          }
52      }
```

The Web Container Model

Need for using RequestDispatcher – Redirect

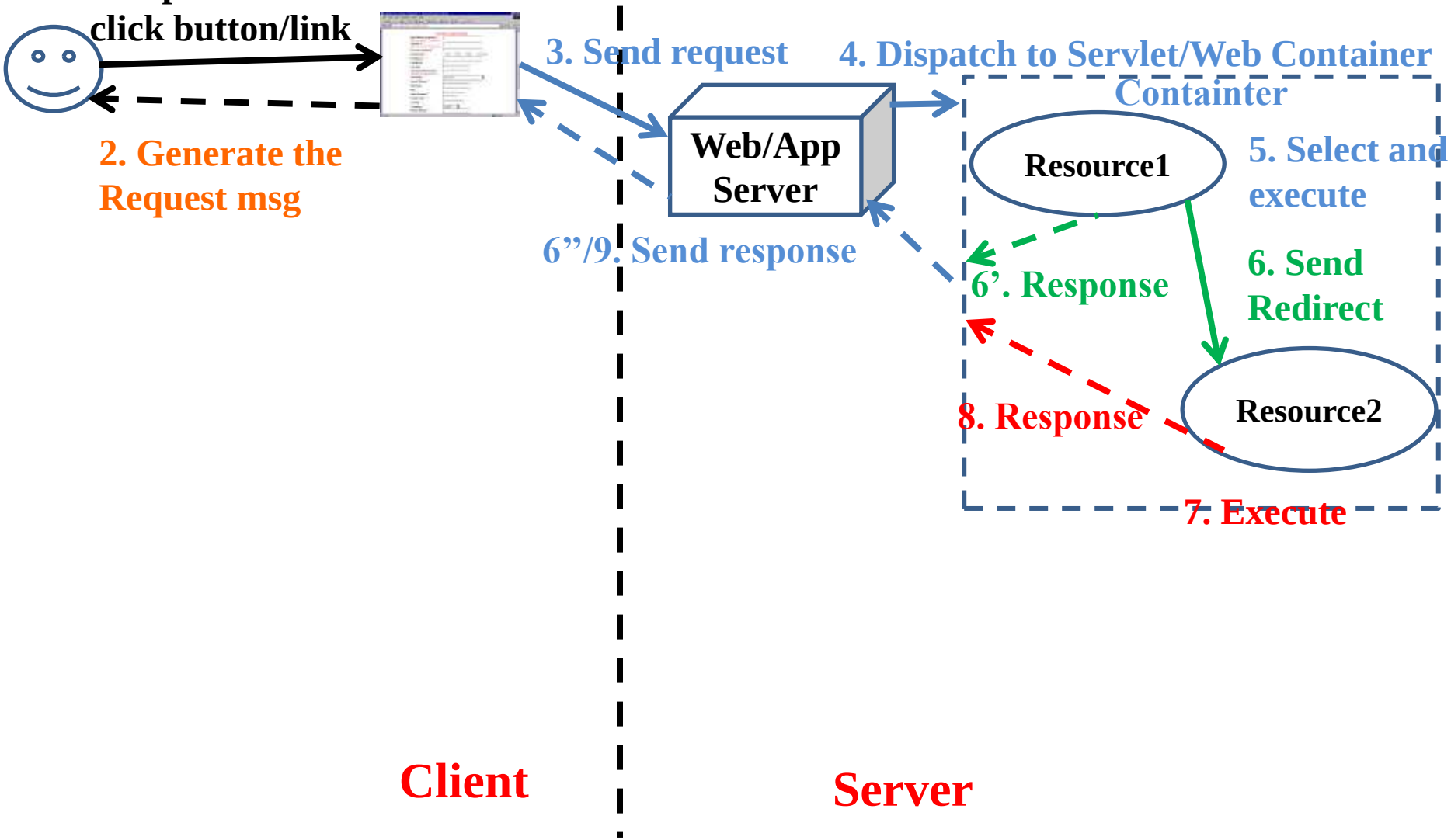




The Web Container Model

Need for using RequestDispatcher

Redirect Mechanism



The Web Container Model

Request Dispatching

- Is a **mechanism** for **controlling** the **flow of control within** the **web resources** in the web application
- The ServletRequest and ServletContext support the **getRequestDispacher(String path) method**
  - **Returns** RequestDispacher instance
  - The path parameter can be a full path beginning at the context root ("/") – **requirement with ServletContext**
  - The ServletContext offers the **getNameDispatcher(String name)** method that requires providing the resource's name to want to execute (e.g. the name must match one of the <servlet-name>)
- A **RequestDispacher object**
  - Is **created** by the **servlet container**
  - **Redirect** the **client request to** a **particular Web page**

The Web Container Model

Using RequestDispatcher

Methods	Descriptions
forward	- **Redirect** the **output** to another servlet - **Forward** the **request** to another Servlet to process the client request. - **Ex**: RequestDispatcher rd = request.getRequestDipatcher(“home.jsp”); rd.forward(request, response);
include	- **Include** the **content** of another servlet into the current output stream - Include the **output** of another Servlet to process the client request - **Ex** RequestDispatcher rd = request.getRequestDipatcher(“home.jsp”); rd.include (request, response);

The Web Container Model

Using RequestDispatcher – Example

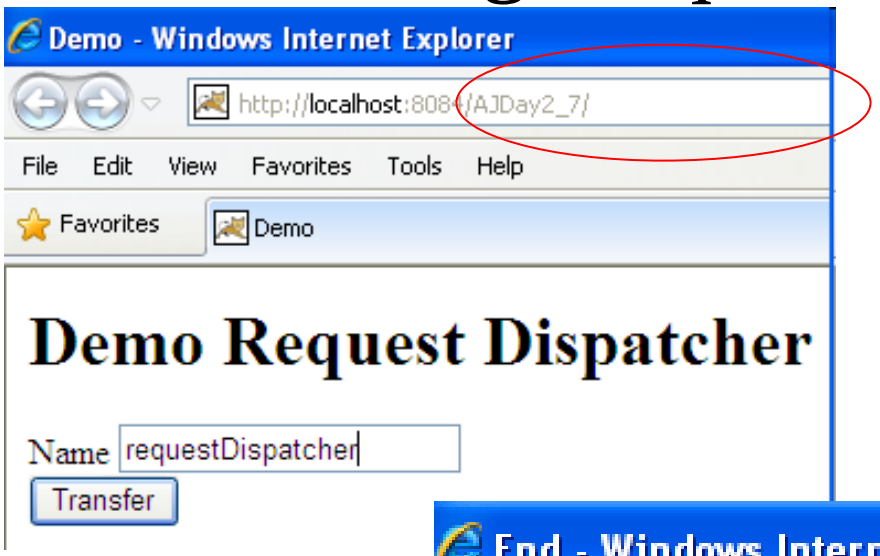
```html
<body>
    <h1>Demo Request Dispatcher</h1>
    <form action="MiddleServlet">
        Name <input type="text" name="txtName" value="" /><br/>
        <input type="submit" value="Transfer" />
    </form>
</body>
```

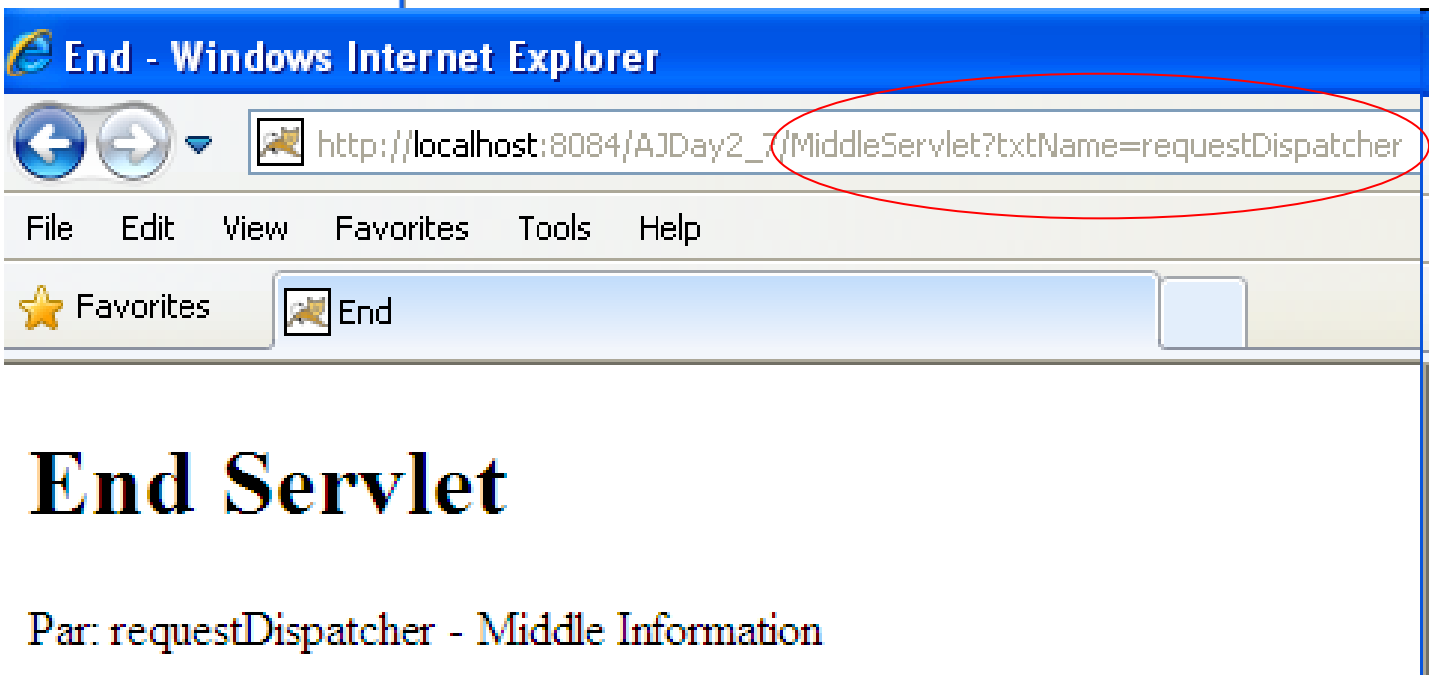
MiddleServlet.java | EndServlet.java

```java
28          */
29          protected void processRequest(HttpServletRequest request, HttpServletRespo
30          throws ServletException, IOException {
31              response.setContentType("text/html;charset=UTF-8");
32              PrintWriter out = response.getWriter();
33              try {
34
35                  out.println("<html>");
36                  out.println("<head>");
37                  out.println("<title>Middle</title>");
38                  out.println("</head>");
39                  out.println("<body>");
40                  out.println("<h1>Middle Servlet</h1>");
41
42                  request.setAttribute("Middle", "Middle Information");
43
44                  RequestDispatcher rd = request.getRequestDispatcher("EndServlet");
45                  rd.forward(request, response);
46
47                  out.println("</body>");
48                  out.println("</html>");
49
50              } finally {
```

The Web Container Model

Using RequestDispatcher – Example

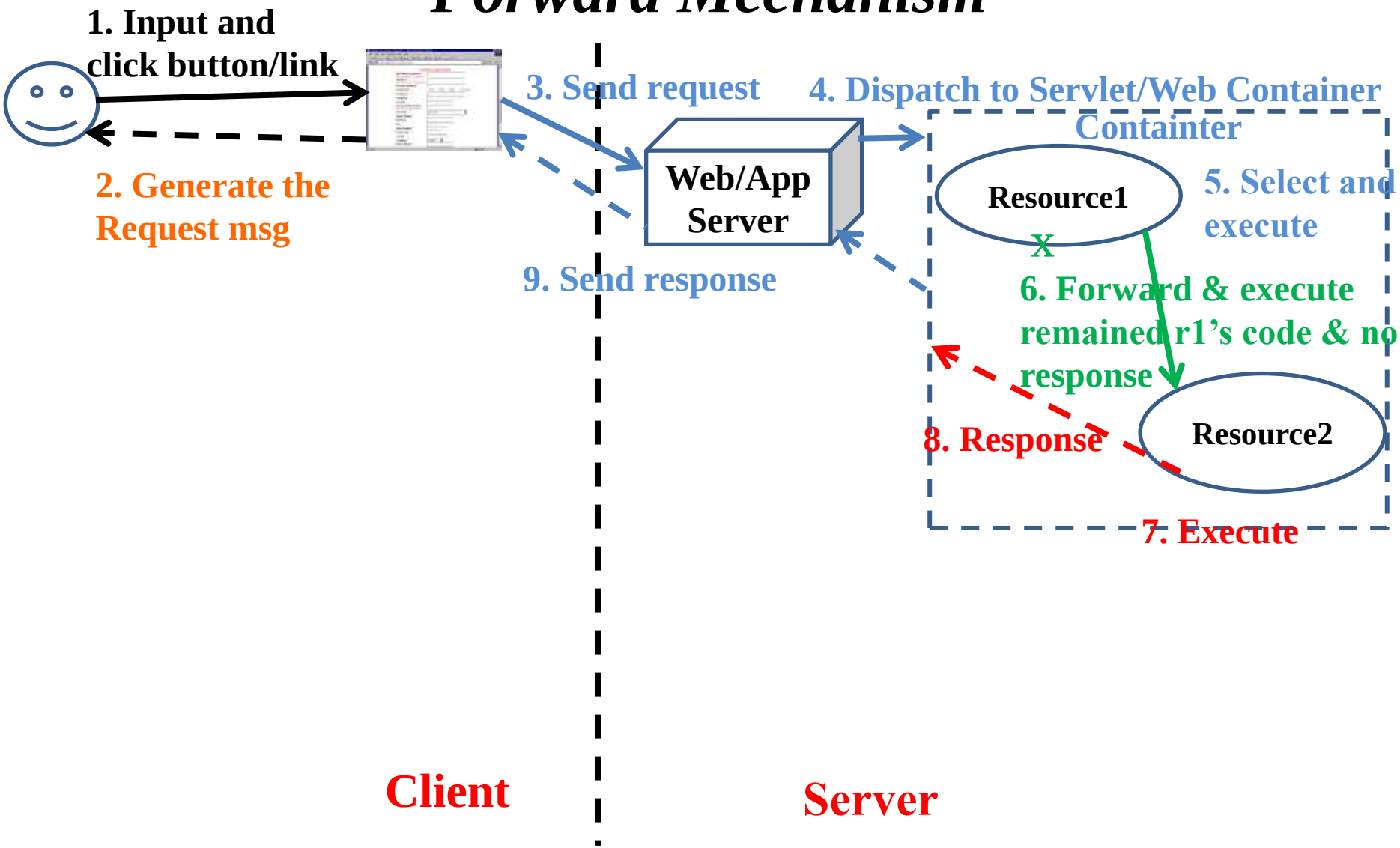




The Web Container Model

Need for using RequestDispatcher

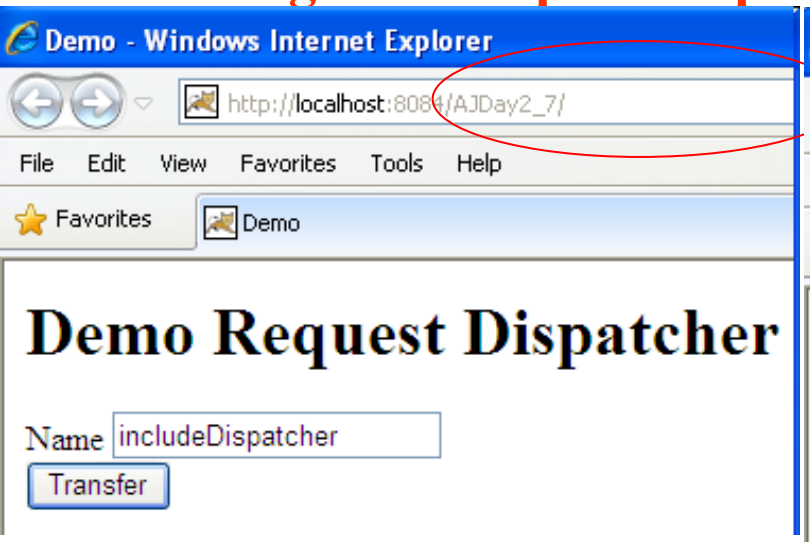
Forward Mechanism

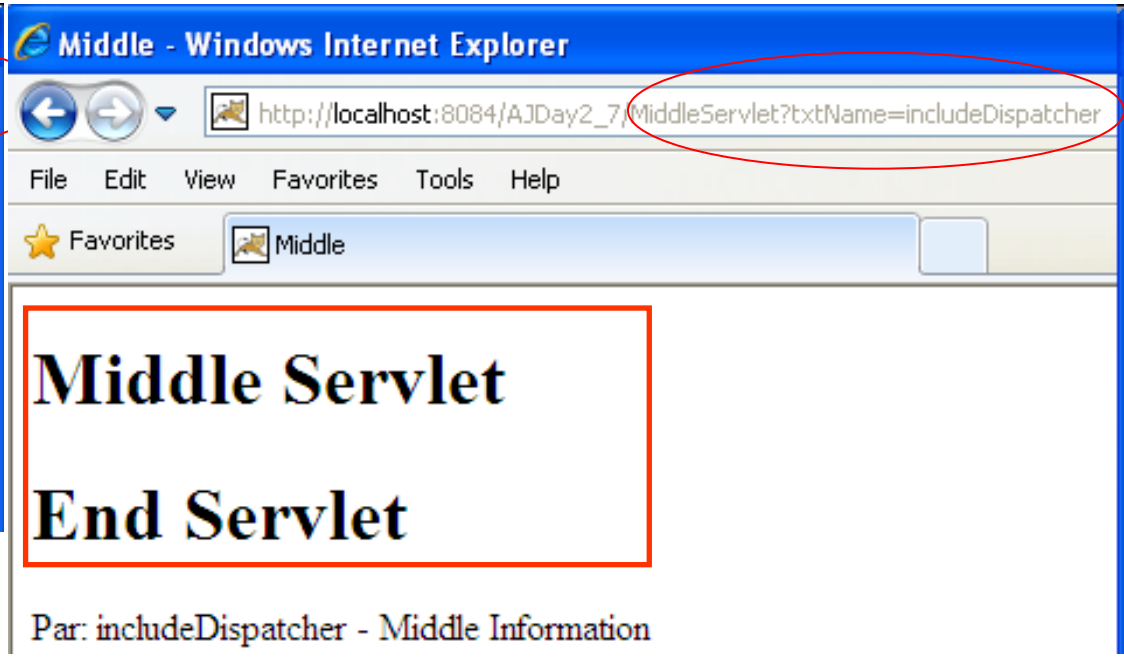


The Web Container Model

Using RequestDispatcher – Example

Change the RequestDispatch – forward method to include method



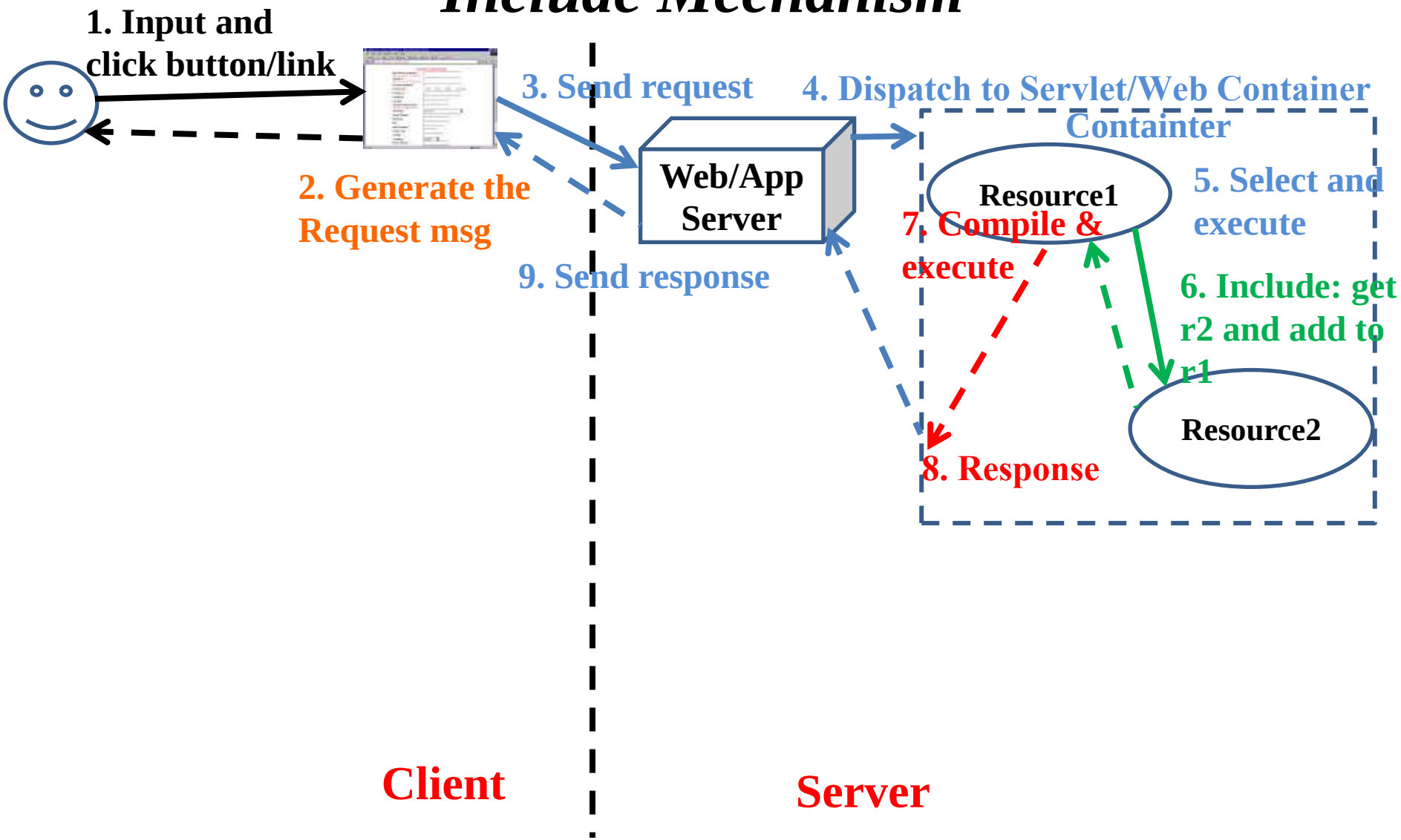



```
request.setAttribute("Middle", "Middle Information");

RequestDispatcher rd = request.getRequestDispatcher("EndServlet");

rd.include(request, response);
out.println("</body>");
out.println("</html>");
```

The Web Container Model

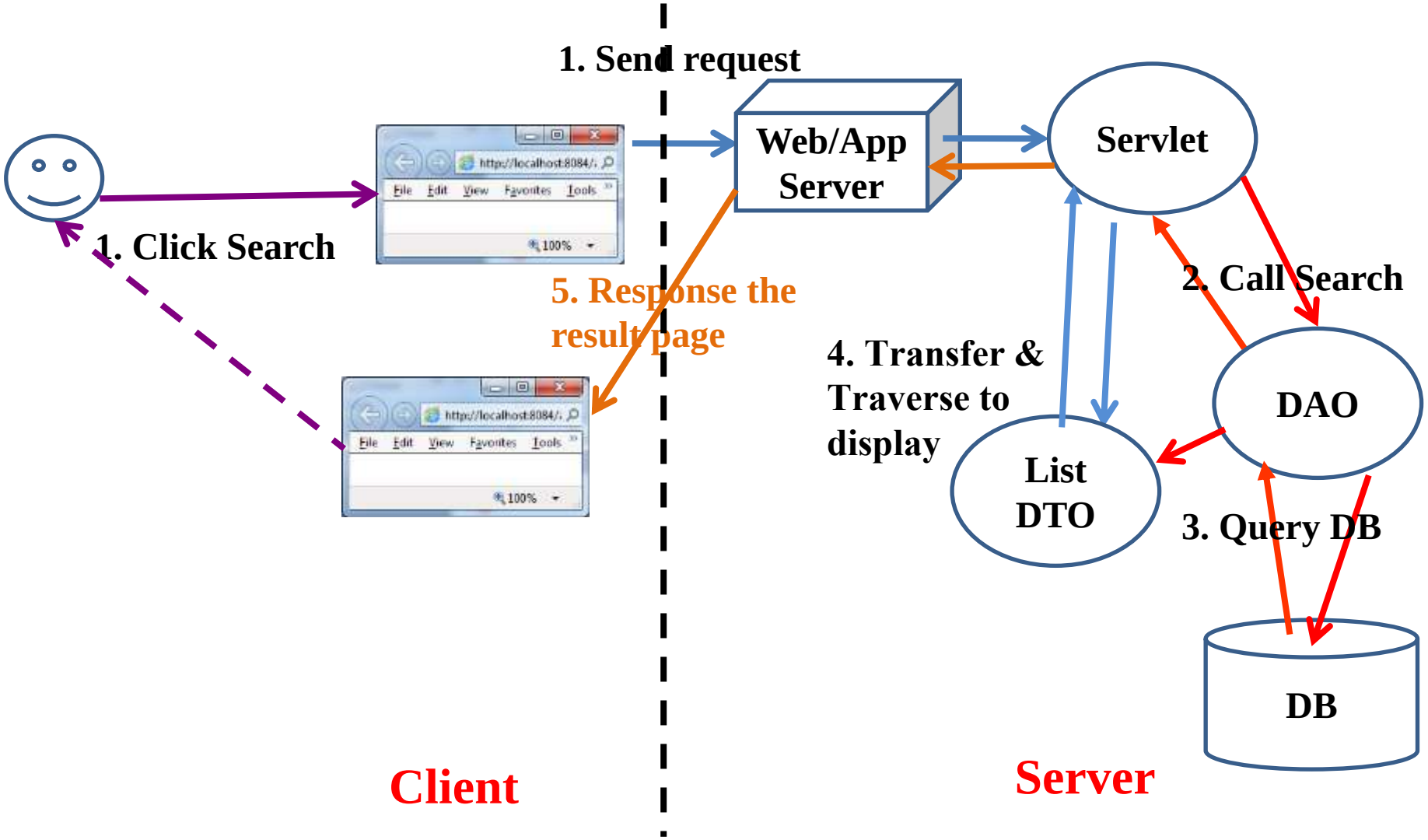
Need for using RequestDispatcher

Include Mechanism



How To Transfer

Interactive Server Model – Implementation



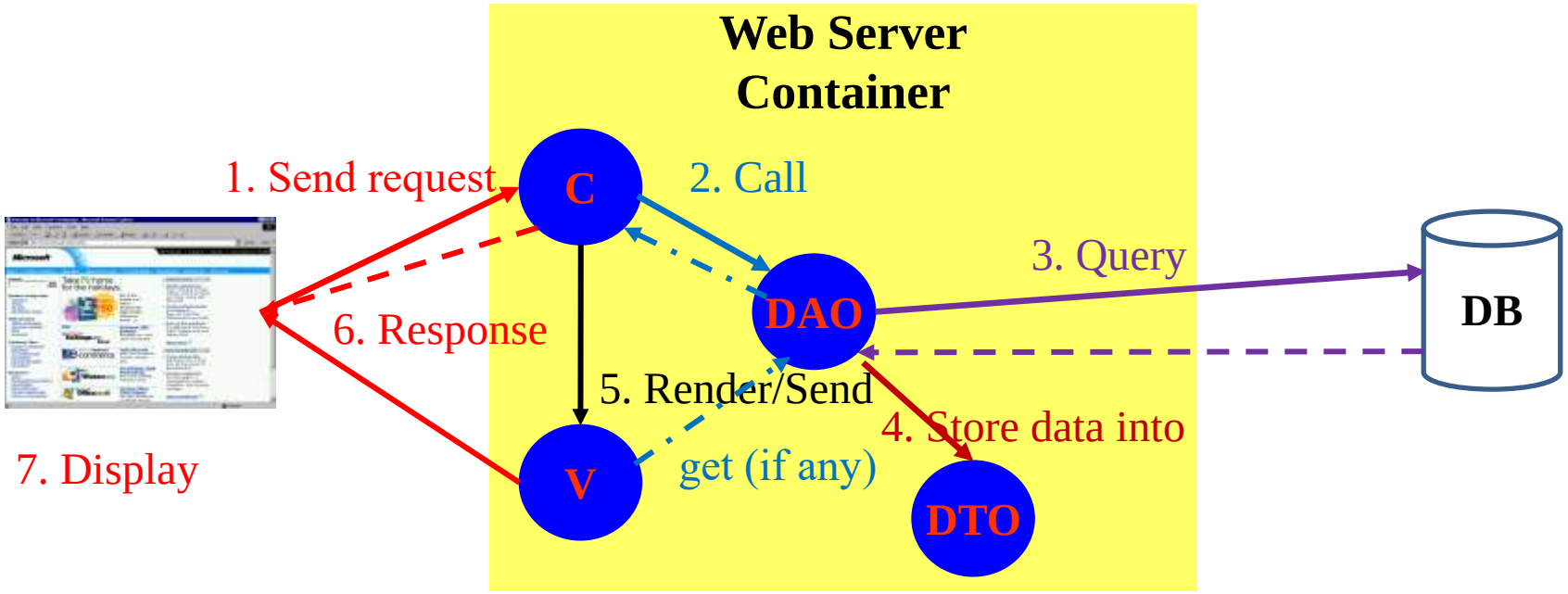
Fpt University

Summary

- **How to deploy the Web Application to Web Server?**
  - Web applications Structure
  - Request Parameters vs. Context Parameters vs. Config/Servlet Parameters
  - Application Segments vs. Scope
- **How to transfer from resources to others with/without data/objects?**
  - Attributes vs. Parameters vs. Variables
  - Redirect vs. RequestDispatcher

Q&A

Summary


Web Server
Container
1. Send request
C
2. Call
3. Query
DAO
DB
6. Response
5. Render/Send
4. Store data into
7. Display
V
get (if any)
DTO

Next Lecture

- **How to upgrade Application in previous topics approach MVC Model**
  - **Using JSP to View**
  - **MVC Pattern Design**

Next Lecture

Web based App - PBL

Web

- Day 1, 2, 3, 4, 5, 6 – Login Servlet, JDBC ☑
- Day 7, 8, 9 – Search Break Down ☑
- Day 10, 11 – MVC2 JSP
- Day 12, 13, 14, 15 – CUD, Shopping Carts Sessions
- Day 16, 17 – Login JavaBeans
- Day 18, 19, 20 – CRUD MVC 2 Complete - JSTL - Taglib
- Day 21, 22, 23 – Filter MVC2 Using Filter as Controller

Appendix – How to Transfer DTO

RegistrationDTO.java

Source History

```
6   package sample.registration;
7   import ...
8   /**
9    *
10   * @author kieukhanh
11   */
12  public class RegistrationDTO implements Serializable {
13      private String username;
14      private String password;
15      private String lastname;
16      private boolean role;
17      public RegistrationDTO() {...2 lines }
19      public RegistrationDTO(String username, String password,
20              String lastname, boolean role) {...6 lines }
26      /**...3 lines */
29      public String getUsername() {...3 lines }
32      /**...3 lines */
35      public void setUsername(String username) {...3 lines }
38      /**...3 lines */
41      public String getPassword() {...3 lines }
44      /**...3 lines */
47      public void setPassword(String password) {...3 lines }
50      /**...3 lines */
53      public String getLastname() {...3 lines }
56      /**...3 lines */
59      public void setLastname(String lastname) {...3 lines }
62      /**...3 lines */
65      public boolean isRole() {...3 lines }
68      /**...3 lines */
71      public void setRole(boolean role) {...3 lines }
74  }
```

How to Transfer DAO

RegistrationDAO.java

Source History

```
20      * @author kieukhanh
21      */
22     public class RegistrationDAO implements Serializable {
23         public boolean checkLogin(String username, String password)
24             throws SQLException, NamingException {...35 lines }
59
60         private List<RegistrationDTO> listAccounts;
61
62         public List<RegistrationDTO> getListAccounts() {
63             return listAccounts;
64         }
65
66         public void searchLastname(String searchValue)
67             throws SQLException, NamingException {
68
69             Connection con = null;
70             PreparedStatement stm = null;
71             ResultSet rs = null;
72
73             try {
74                 con = DBUtils.makeConnection();
75
76                 if (con != null) {
77                     String sql = "Select * From Registration Where lastname Like ?";
```

How to Transfer DAO

```
79                  stm = con.prepareStatement(sql);
80                  stm.setString(1, "%" + searchValue + "%");
81
82                  rs = stm.executeQuery();
83                  while (rs.next()) {
84                      String username = rs.getString("username");
85                      String password = rs.getString("password");
86                      String lastname = rs.getString("lastname");
87                      boolean role = rs.getBoolean("isAdmin");
88
89                      RegistrationDTO dto =
90                              new RegistrationDTO(username, password, lastname, role);
91
92                      if (this.listAccounts == null) {
93                          this.listAccounts = new ArrayList<RegistrationDTO>();
94                      }
95
96                      this.listAccounts.add(dto);
97                  }
98              }
99          } finally {
100             if (rs != null) {
101                 rs.close();
102             }
103             if (stm != null) {
104                 stm.close();
105             }
106             if (con != null) {
107                 con.close();
108             }
```

How to Transfer

Controller Servlet

SE1162Servlet.java

Source History

```
 * @author kieukhanh
 */
public class SE1162Servlet extends HttpServlet {
    private final String loginServlet = "LoginServlet";
    private final String loginPage = "login.html";
    private final String searchServlet = "SearchServlet";

    /** Processes requests for both HTTP <code>GET</code> and <code>POST</co
    protected void processRequest(HttpServletRequest request, HttpServletResp
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        String button = request.getParameter("btAction");
        String url = loginPage;
        try {
            if (button == null) {

            } else if (button.equals("Login")) {
                url = loginServlet;
            } else if (button.equals("Search")) {
                url = searchServlet;
            }
        } finally {
            RequestDispatcher rd = request.getRequestDispatcher(url);
            rd.forward(request, response);

            out.close();
        }
    }
```

How to Transfer

Login Servlet

LoginServlet.java

Source History

```
 * @author kieukhanh
 */
public class LoginServlet extends HttpServlet {

    private final String searchPage = "search.html";
    private final String invalidPage = "invalid.html";

    /** Processes requests for both HTTP <code>GET</code> and <code>
    protected void processRequest(HttpServletRequest request, HttpSer
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        try {
            String username = request.getParameter("txtUsername");
            String password = request.getParameter("txtPassword");

            RegistrationDAO dao = new RegistrationDAO();
            boolean result = dao.checkLogin(username, password);

            String url = invalidPage;

            if (result) {
                url = searchPage;
            }
            response.sendRedirect(url);
        } catch (NamingException ex) {
            ex.printStackTrace();
        } catch (SQLException ex) {
            ex.printStackTrace();
        } finally {
```

How to Transfer

Search Servlet

SearchServlet.java

Source History

```
24      * @author kieukhanh
25      */
26     @WebServlet(name = "SearchServlet", urlPatterns = {"/SearchServlet"})
27     public class SearchServlet extends HttpServlet {
28         private final String searchPage = "search.html";
29         private final String showSearchResult = "ShowSearchResultServlet";
30         /** Processes requests for both HTTP <code>GET</code> and <code>POST
39         protected void processRequest(HttpServletRequest request, HttpServlet
40                 throws ServletException, IOException {
41             response.setContentType("text/html;charset=UTF-8");
42             PrintWriter out = response.getWriter();
43             String url = searchPage;
44             String searchValue = request.getParameter("txtSearchValue");
45             try {
46                 if (!searchValue.isEmpty()) {
47                     RegistrationDAO dao = new RegistrationDAO();
48                     dao.searchLastname(searchValue);
49                     List<RegistrationDTO> result = dao.getListAccounts();
50                     request.setAttribute("SEARCHRESULT", result);
51
52                     url = showSearchResult;
53                 }
54             } catch (NamingException ex) {
                   ex.printStackTrace();
56             } catch (SQLException ex) {
                   ex.printStackTrace();
58             } finally {
59                 RequestDispatcher rd = request.getRequestDispatcher(url);
60                 rd.forward(request, response);
61                 out.close();
```

How to Transfer

Search Result Servlet

ShowSearchResultServlet.java

Source History

```
19      *
20      * @author kieukhanh
21      */
22     @WebServlet(name = "ShowSearchResultServlet", urlPatterns = {"/ShowSearchResultSer
23     public class ShowSearchResultServlet extends HttpServlet {
24
25         /** Processes requests for both HTTP <code>GET</code> and <code>POST</code> ..
34         protected void processRequest(HttpServletRequest request, HttpServletResponse 
35                 throws ServletException, IOException {
36             response.setContentType("text/html;charset=UTF-8");
37             PrintWriter out = response.getWriter();
38             try {
39                 /* TODO output your page here. You may use following sample code. */
40                 out.println("<!DOCTYPE html>");
41                 out.println("<html>");
42                 out.println("<head>");
43                 out.println("<title>Search Result</title>");
44                 out.println("</head>");
45                 out.println("<body>");
46                 out.println("<h1>Search Result</h1>");
47
48                 String searchValue = request.getParameter("txtSearchValue");
49
50                 out.println("Your search value is " + searchValue);
51
52                 List<RegistrationDTO> result =
                           (List<RegistrationDTO>) request.getAttribute("SEARCHRESULT");
54
```

How to Transfer

Search Result Servlet

ShowSearchResultServlet.java

Source | History

```
54
55                  if (result != null) {
56                      out.println("<table border='1'>");
57                      out.println("<thead>");
58                      out.println("<tr>");
59                      out.println("<th>No.</th>");
60                      out.println("<th>Username</th>");
61                      out.println("<th>Password</th>");
62                      out.println("<th>Lastname</th>");
63                      out.println("<th>Role</th>");
64                      out.println("</tr>");
65                      out.println("</thead>");
66                      out.println("<tbody>");
67                      int count = 0;
68                      for (RegistrationDTO dto : result) {
69                          out.println("<tr>");
70                          out.println("<td>"
71                                  + ++count
72                                  + ".</td>");
73                          out.println("<td>"
74                                  + dto.getUsername()
75                                  + "</td>");
76                          out.println("<td>"
77                                  + dto.getPassword()
78                                  + "</td>");
79                          out.println("<td>"
80                                  + dto.getLastname()
81                                  + "</td>");
82                          out.println("<td>"
83                                  + dto.isRole()
84                                  + "</td>");
```

```
85                          out.println("</tr>");
86                      }
87
88                      out.println("</tbody>");
89                      out.println("</table>");
90                  } else {
91                      out.println("<h2>No record is matched</h2>");
92                  }
93
94                  out.println("</body>");
95                  out.println("</html>");
96              } finally {
97                  out.close();
98              }
99          }
```